Q1 Two dice are tossed and their sum, S, is calculated.

(i) Find the probability that S is 12.

(ii) Find the probability that S is less than 7.

(iii) It is known that a 4 appears on at least one of the dice. Find the probability that S is at least 8.

NOTE: **The same answer may be used for more than one question.**

Q2 Two dice are tossed and their product, P, is calculated.

(i) Find the probability that P is 6.

(ii) Find the probability that P is even.

(iii) Find the probability that scores of 2, 2 and 2 are recorded in three tosses of the dice.

Q3 The digits 1, 2, 4, 6 are drawn at random without replacement to form a four-digit number.

(i) Find the probability that the number is less than 3000.

(ii) Find the probability that the number is more than 2500.

(iii) Find the probability that the number is odd.

Q4 In a class of 32 students, 18 study History, 24 study Geography and 2 study neither. What is the probability that:

(i) a randomly chosen student studies both History and Geography?

(ii) a randomly chosen student studies Geography but not History?

(iii) two randomly chosen students both study History?

Q5 20 dogs were studied to see which of 3 characteristics A, B or C were displayed. All of the dogs displayed at least one of the characteristics, some displayed two, while others displayed all three.

In total:

- 5 dogs displayed all 3 characteristics
- 7 dogs displayed characteristics A and B
- 14 dogs displayed characteristic A
- 3 dogs displayed characteristics A and C but not B
- 8 dogs displayed characteristic C

(i) Find the probability that a randomly chosen dog displayed characteristic A.

(ii) Find the probability that two randomly chosen dogs both displayed characteristic B.

ANSWERS

A) $\frac{1}{36}$

B) $\frac{1}{9}$

C) $\frac{1}{1728}$

D) $\frac{153}{496}$

E) $\frac{1}{18}$

F) $\frac{37}{95}$

G) $\frac{3}{4}$

H) $\frac{1}{12}$

I) $\frac{1}{54}$

J) $\frac{1}{5832}$

K) $\frac{17}{24}$

L) $\frac{3}{8}$

M) $\frac{13}{24}$

N) $\frac{1}{4}$

O) $\frac{5}{12}$

P) $\frac{5}{8}$

Q) $\frac{81}{256}$

R) $\frac{1}{2}$

S) $\frac{169}{400}$

T) $\frac{39}{95}$

U) $\frac{7}{12}$

V) $\frac{3}{11}$

W) $\frac{9}{10}$

X) $\frac{5}{36}$

Y) $\frac{5}{11}$

Z) $\frac{7}{10}$

Q1

11	21	31	41	51	61
12	22	32	42	52	62
13	23	33	43	53	63
14	24	34	44	54	64
15	25	35	45	55	65
16	26	36	46	56	66

(i) $P(S=12)=\frac{1}{36}$

(ii) $P(S<7)=\frac{15}{36}=\frac{5}{12}$

(iii) If a 4 appears, the possible results are $\{14,24,34,44,54,64,41,42,43,45,46\}$

$\therefore P(S\geq 8)=\frac{5}{11}$

Q2 P, **the product**, is written in brackets ( )

1,1 (1)	2,1(2)	3,1(3)	4,1(4)	5,1 (5)	6,1 (6)
1,2(2)	2,2(4)	3,2(6)	4,2(8)	5,2(10)	6,2(12)
1,3(3)	2,3(6)	3,3(9)	4,3(12)	5,3(15)	6,3(18)
1,4(4)	2,4(8)	3,4(12)	4,4(16)	5,4(20)	6,4(24)
1,5(5)	2,5(10)	3,5(15)	4,5(20)	5,5(25)	6,5(30)
1,6(6)	2,6(12)	3,6(18)	4,6(24)	5,6(30)	6,6(36)

(i) $P(6)=\frac{4}{36}=\frac{1}{9}$

(ii) $P(\text{even})=\frac{27}{36}=\frac{3}{4}$

(iii) $P(2)=\frac{2}{36}=\frac{1}{18}$

$$P(2,2,2)=\frac{1}{18}\times\frac{1}{18}\times\frac{1}{18}=\frac{1}{5832}$$

Q3 LIST $\Rightarrow$ using 1, 2, 4, 6

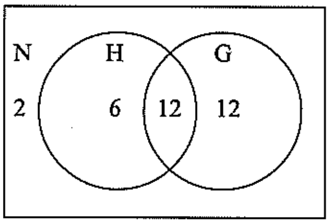
1246	2146	4126	6124
1264	2164	4162	6142
1426	2416	4216	6214
1462	2461	4261	6241
1624	2614	4612	6412
1642	2641	4621	6421

(i) $P(<3000)=\frac{12}{24}=\frac{1}{2}$

(ii) $P(>2500)=\frac{14}{24}=\frac{7}{12}$

(iii) $P(\text{odd})=\frac{6}{24}=\frac{1}{4}$

Q4

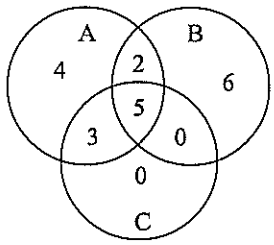


(i) $P(H \text{ and } G)=\frac{12}{32}=\frac{3}{8}$

(ii) $P(G, \text{not } H)=\frac{12}{32}=\frac{3}{8}$

(iii) $P(H,H)=\frac{18}{32}\times\frac{17}{31}=\frac{153}{496}$

Q5



(i) $P(A)=\frac{14}{20}=\frac{7}{10}$

(ii) $P(B,B)=\frac{13}{20}\times\frac{12}{19}=\frac{39}{95}$

Q1 *"The government states that one in every 12 tickets wins a prize in the state lotteries. So far I have bought 11 tickets without success. I am therefore certain to win the next time I buy a ticket."*

Comment on the validity of this claim.

Q2 Comment on the accuracy of each statement.

(a) "If a page is chosen at random from all pages of a book, then the probability that the *first* digit of its page number is 1 will be about $\frac{1}{10}$."

(b) "If a page is chosen at random from all pages of a book, then the probability that the *last* digit of its page number is 1 will be about $\frac{1}{10}$."

Q3 The probability of a dog having triplets is 0.4. Two of Mark's dogs are expecting. What is the probability that:

(a) both dogs will have triplets?

(b) only one dog has triplets?

(c) at least one of the dogs has triplets?

Q4 In a large country town, 58% are male and 42% female.
If 3 people are chosen at random, find the probability that:

(a) all 3 are female.

(b) exactly one is male.

(c) at least one is male.

Q5 A factory assembles toys. Each toy requires two major parts, A and B, to be fitted together in the final stage.

It is known that 5% of all the A parts are defective and 7% of all the B parts are defective.

Find the probability that a toy chosen at random is not defective.

Q6 It has been estimated that the probability of a dog living beyond the age of 7 is 0.8, and that of a cat living beyond the age of 7 is 0.9. Lydia owns a cat and Kath owns a dog.

Using a tree diagram calculate the probability that:

(a) both animals live past 7 years.

(b) the dog lives past 7 years but not the cat.

(c) at least one lives past 7 years.

Q7 Debbie is playing in a tennis round-robin. She plays 3 matches and needs to win at least two of these to progress to the semi-finals. She has a probability of 0.6 of winning her first match. If she wins a match, her confidence increases such that the probability of her winning the next match is 0.8. Conversely, if she loses a match, the chance that she wins the next drops to 0.3.

Using a tree diagram, calculate the probability that Debbie:

(a) wins all 3 matches.

(b) loses all 3 matches.

(c) wins at least one match.

(d) progresses to the semi-finals.

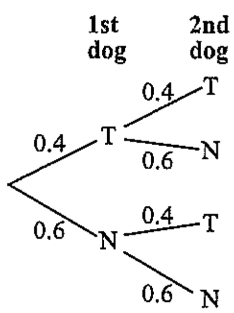
Q1 This statement is FALSE, because each time a ticket is bought, it has a $\frac{1}{12}$ chance of winning a prize, regardless of what has happened previously.

2(a) This statement is FALSE. Consider a book of say 200 pages. You would find that there are 110 pages where 1 is the first digit of the page number.

Hence the probability would generally be far greater than $\frac{1}{10}$.

2(b) This statement is pretty close to being TRUE. In sequence, the page numbers would finish as 0,1,2,3,4,5,6,7,8,9.
This is repeated until the end of the book.
Hence the probability that the page number ends in 1 is very close to 1 in 10.

Q3 P(Triplets) = 0.4, P(Not triplets) = 0.6



3(a) $P(TT) = 0.4 \times 0.4$
$= 0.16$

3(b) P(TN) or P(NT)
$= (0.4 \times 0.6) + (0.6 \times 0.4)$
$= 0.48$

3(c) $P(\text{at least 1 set of triplets}) = 1 - P(NN)$
$= 1 - (0.6 \times 0.6)$
$= 0.64$

Q4 $P(M) = 0.58$, $P(F) = 0.42$

4(a) $P(FFF) = 0.42 \times 0.42 \times 0.42$
$= 0.074$
$= 7.4\%$

4(b) P(1 male) = P(MFF) *or* P(FMF) *or* P(FFM)
$= (0.58 \times 0.42 \times 0.42)$
$+ (0.42 \times 0.58 \times 0.42)$
$+ (0.42 \times 0.42 \times 0.58)$
$= 0.307$
$= 30.7\%$

4(c) $P(\text{at least 1 male}) = 1 - P(FFF)$
$= 1 - 0.074$ ......from (a)
$= 0.926$
$= 92.6\%$

Q5 A: P(Defective) = 0.05, P(Good) = 0.95

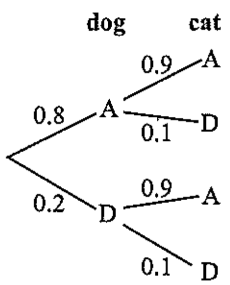
B: P(Defective) = 0.07, P(Good) = 0.93

$\therefore$ P(GoodA, GoodB) $= 0.95 \times 0.93$
$= 0.8835$

Q6

Dog: P(Alive) = 0.8
P(Dead) = 0.2

Cat: P(Alive) = 0.9
P(Dead) = 0.1

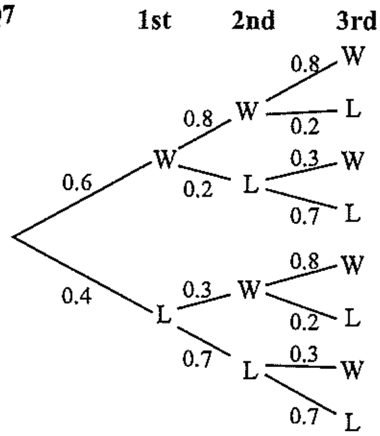


6(a) $P(LL) = 0.8 \times 0.9$
$= 0.72$

6(b) $P(LA) = 0.8 \times 0.1$
$= 0.08$

6(c) $P(\text{at least 1 lives}) = 1 - P(DD)$
$= 1 - (0.2 \times 0.1)$
$= 0.98$

Q7



7(a) $P(WWW) = 0.6 \times 0.8 \times 0.8$
$= 0.384$

7(b) $P(LLL) = 0.4 \times 0.7 \times 0.7$
$= 0.196$

7(c) $P(\text{wins at least 1 match}) = 1 - P(LLL)$
$= 1 - 0.196$
$= 0.804$

7(d) To progress to semi-finals, Debbie needs to win 2 or more matches.

$$\therefore P(\text{semi-final}) = P(WWW) \text{ or } P(WWL) \text{ or } P(WLW) \text{ or } P(LWW)$$
$$= (0.6 \times 0.8 \times 0.8) + (0.6 \times 0.8 \times 0.2) + (0.6 \times 0.2 \times 0.3) + (0.4 \times 0.3 \times 0.8)$$
$$= 0.612$$

Q1 Andrew buys 5 tickets in a raffle in which 100 tickets are sold. Two different tickets are to be drawn out for first and second prize.

Find the probability that Andrew:

(i) wins both prizes. (ii) wins one prize. (iii) wins at least one prize.

Q2 Sarah earns 4 tickets in a class raffle for excellent work. The teacher gives out 50 tickets in total and draws three different tickets for 1st, 2nd and 3rd prizes.

Find the probability that Sarah:

(i) wins all three prizes.

(ii) does not win a prize.

(iii) wins 2nd prize only.

(iv) wins at least one prize.

Q3 A bag contains 8 green balls and 4 blue balls. Emilie selects three balls at random, one at a time and without replacement.

Find the probability that Emilie selects:

(i) 3 green balls.

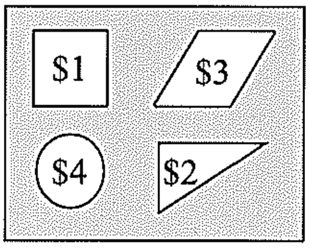
(ii) GREEN, BLUE, BLUE in that order.

(iii) at least one blue ball.

NOTE: **The same answer may be used for more than one question.**

Q4 The diagram shows a rectangular board, 70 cm by 55 cm. It has 4 shapes painted on it:

- a square of side 20 cm;
- a parallelogram with base length 18 cm and height 20 cm;
- a circle of radius 9 cm; and
- a triangle with base length 24 cm and height 22 cm.



A game consists of throwing 3 darts, one at a time, at the board. If the square is hit, $1 is won, the parallelogram hit will win you $3, the circle $4 and the triangle $2.

(Any throw missing the board completely is not counted as a valid throw. Hence it is thrown again until a valid throw is recorded.)

On any one valid throw of the dart, find the probability, to 3 decimal places, that:

(i) the square is hit.

(ii) the circle is hit.

(iii) the parallelogram is hit.

(iv) the triangle is hit.

(v) In one game (3 valid throws recorded), find the probability of winning exactly $10.

Answers

A) 0.0026
B) 0.0035
C) 0.055
D) 0.066
E) 0.069
F) 0.094
G) 0.098
H) 0.104
I) $\frac{41}{55}$
J) $\frac{221}{980}$
K) $\frac{23}{198}$
L) $\frac{97}{990}$
M) $\frac{4}{55}$
N) $\frac{659}{980}$
O) $\frac{1}{495}$
P) $\frac{14}{55}$
Q) $\frac{241}{980}$
R) $\frac{5}{4900}$
S) $\frac{19}{198}$
T) $\frac{31}{55}$
U) $\frac{71}{980}$
V) $\frac{759}{980}$
W) $\frac{1}{4900}$
X) $\frac{12}{55}$
Y) $\frac{7}{495}$
Z) $\frac{69}{980}$

1(i) $P(WW) = \frac{5}{100} \times \frac{4}{99} = \frac{1}{495}$

1(ii) P(WL) or P(LW)

$$= \left(\frac{5}{100} \times \frac{95}{99}\right) + \left(\frac{95}{100} \times \frac{5}{99}\right)$$

$$= \frac{19}{198}$$

1(iii) $P(\text{at least 1 prize}) = 1 - P(LL)$

$$= 1 - \left(\frac{95}{100} \times \frac{94}{99}\right)$$

$$= \frac{97}{990}$$

2(i) $P(WWW) = \frac{4}{50} \times \frac{3}{49} \times \frac{2}{48}$

$$= \frac{3}{14\,700}$$

2(ii) $P(LLL) = \frac{46}{50} \times \frac{45}{49} \times \frac{44}{48}$

$$= \frac{759}{980}$$

2(iii) $P(LWL) = \frac{46}{50} \times \frac{4}{49} \times \frac{45}{48}$

$$= \frac{69}{980}$$

2(iv) $P(\text{at least 1 prize}) = 1 - P(LLL)$

$$= 1 - \frac{759}{980}$$

$$= \frac{521}{980}$$

3(i) $P(GGG) = \frac{8}{12} \times \frac{7}{11} \times \frac{6}{10} = \frac{14}{55}$

3(ii) $P(GBB) = \frac{8}{12} \times \frac{4}{11} \times \frac{3}{10}$

$$= \frac{4}{55}$$

3(iii) $P(\text{at least 1 blue}) = 1 - P(GGG)$

$$= 1 - \frac{14}{55}$$

$$= \frac{41}{55}$$

Q4 Probabilities are calculated according to their areas.

Full rectangular board $\Rightarrow$ AREA = 3850

$A_{\square} = 20 \times 20 = 400$

$A_{\bigcirc} = \pi \times 9^2 = 81\pi$

$A_{\text{▱}} = 18 \times 20 = 360$

$A_{\triangledown} = \frac{1}{2} \times 24 \times 22 = 264$

(i) (i) $P(\square) = \frac{400}{3850} = 0.104$ (\$1)

(ii) $P(\bigcirc) = \frac{81\pi}{3850} = 0.066$ (\$4)

(iii) $P(\text{▱}) = \frac{360}{3850} = 0.094$ (\$3)

(iv) $P(\triangledown) = \frac{264}{3850} = 0.069$ (\$2)

(ii) To win \$10, we could get either
A: (two \$4 and one \$2 *in any order*)
or B: (two \$3 and one \$4 *in any order*)

A: P(4, 4, 2) or P(4, 2, 4) or P(2, 4, 4)

$$= 3 \times \frac{81\pi}{3850} \times \frac{81\pi}{3850} \times \frac{264}{8350}$$

$$= 0.000898695....$$

B: P(3, 3, 4) or P(3, 4, 3) or P(4, 3, 3)

$$= 3 \times \frac{360}{3850} \times \frac{360}{3850} \times \frac{81\pi}{3850}$$

$$= 0.00173372....$$

$\therefore P(\$10) = 0.0026$ *(after adding A and B)*